Probe-Gated Critic Transfer

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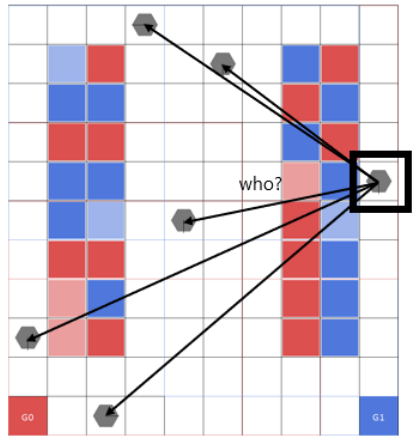
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Outline

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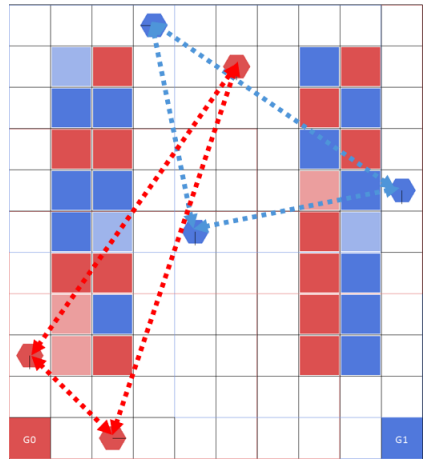
Introduction: Research Background

- **Multi-Agent Reinforcement Learning**: Multi-Agent interact in a single environment and learn their policies
- **Decentralized Learning**: Each agent learns through its observations, actions, and rewards
- **Unknown Team Relations**: Agents aren't given team relationships each other
- **Difficulty of Identifying Teammates**: Even the same team, agents take different actions by their position and role



Introduction: Research Goal

- **Objective**: Discover latent teammates in fully decentralized MARL without team labels
- **Key Idea**: Identify potential teammates by comparing reward to common public-transition probes
- **Contributions**
  - Discover latent teammates through individual reward responses without team labels, using common probes and repeated comparisons.
  - Use identified teammates' Critics through Max-GAE while preserving independent Critic learning.
  - Evaluate online team identification, reward-response separation, and selective Critic access under decentralized learning.

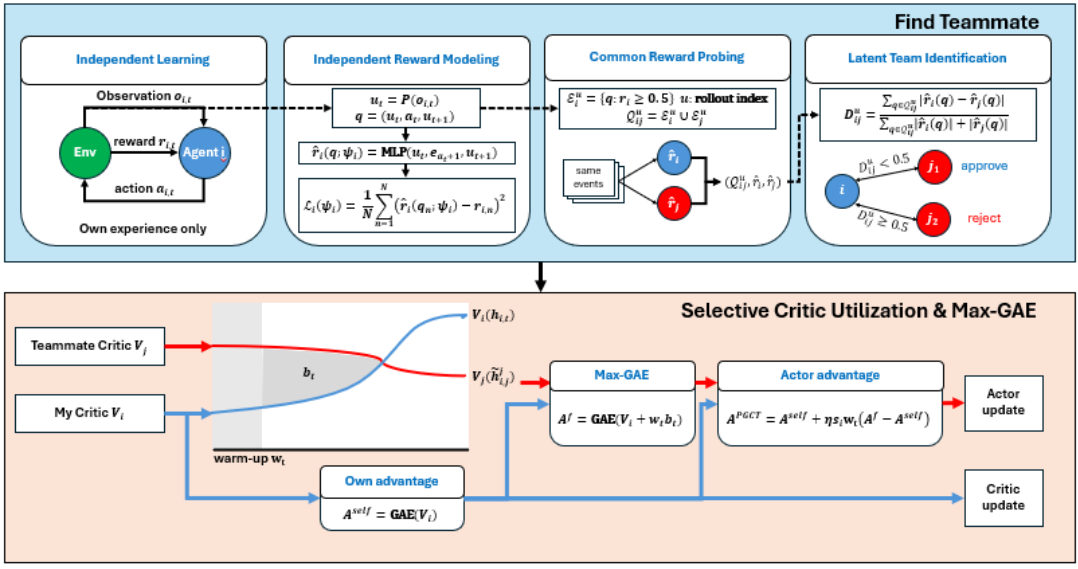


Introduction: Related Works

Approach	Main Idea	Limitation for Our Setting
SEAC (NeurIPS 2020)	Improves learning efficiency through off-policy experience sharing	Does not identify teammates before sharing information
SePS (ICML 2021)	Groups agents by abilities and goals for parameter sharing	Does not compare individual reward responses for identification
Fully Decentralized MARL (ICML 2018)	Learns cooperatively through consensus among networked critics	Does not address unknown team relations
MAMBA (NeurIPS 2020)	Improves policies by combining multiple oracle value functions	Assumes access to oracle values, not learned peer critics

- **Research Gap**: Discover latent teammates from individual reward responses and selectively utilize their critics without predefined team labels

Proposed Method: Overall architecture



Proposed Method: Reward Modeling

- Public transition representation

$$u_{i,t} = P(o_{i,t}), \quad q_{i,t} = (u_{i,t}, a_{i,t}, u_{i,t+1})$$

- Agent-specific reward predictor

$$\hat{r}_i(q; \psi_i) = \mathrm{MLP}_{\psi_i}(u_t, e_{a_t+1}, u_{t+1})$$

- Learning from individual reward targets

$$\mathcal{L}_)(\psi_i) = \frac{1}{N} \sum_{n=1}^{N_i} (\hat{r}_i(q_{i,n}; \psi_i) - r_{i,n})^2$$

Proposed Method: Reward Probe Construction

- **Observed event selection**

$$\mathcal{E}_i^u = \{q_{i,t} : r_i \geq 0.5,\ t \in u\}$$

- **Pairwise common probe set**

$$\mathcal{Q}_{ij}^u = \mathcal{E}_i^u \cup \mathcal{E}_j^u \quad \longrightarrow (\mathcal{Q}_{ij}^u, \hat{r}_i, \hat{r}_j)$$

- **Response Distance and Team Discovery**

$$D_{ij}^u = \frac{\sum_{q \in \mathcal{Q}_{ij}^u} |\hat{r}_i^u(q) - \hat{r}_j^u(q)|}{\sum_{q \in \mathcal{Q}_{ij}^u} (|\hat{r}_i^u(q)| + |\hat{r}_j^u(q)|)}$$

$$\bar{D}_{ij} \longleftarrow (1-\beta)\bar{D}_{ij} + \beta D_{ij}^u$$

$$B_{ij} = \mathbb{1}[i \neq j, n_{ij} \geq 3, \bar{D}_{ij} < 0.5, m_i^u = m_j^u = 1]$$

Proposed Method: Selective Critic Utilization & Max-GAE

$$C_{ij}^{i} = \mathbb{1}[\exists t \in u : ||p_{i,t} - p_{j,t}|| \le r_c]$$
$$\mathcal{G}_i^u = \left\{ j : B_{ij}^u = 1,\ C_{ij}^u = 1,\ m_j^u = 1 \right\}$$

- **Optimistic teammate value uplift**

$$b_{i,t} = \left[\max_{j \in \mathcal{G}_i^u} V_j(\tilde{h}_{i,t}^j) - V_i(h_{i,t}) \right]_+$$
$$f_{i,t} = V_i(h_{i,t}) + w_t b_{i,t}, \qquad A_{i,t}^f = \mathrm{GAE}(f_i)$$

- **Actor advantage (bounded auxiliary update)**

$$A_{i,t}^{\mathrm{PGCT}} = A_{i,t}^{\mathrm{self}} + \eta s_i w_t \left(A_{i,t}^f - A_{i,t}^{\mathrm{self}} \right)$$

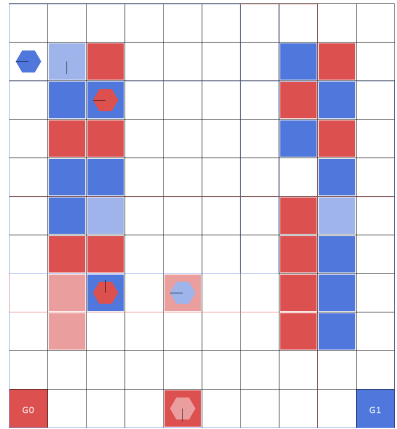
$$s_i = \min \left(1, \frac{\mathrm{std}(A_i^{\mathrm{self}})}{\mathrm{std}(w_t \left(A_{i,t}^f - A_{i,t}^{\mathrm{self}} \right)), 10^{-8}} \right)$$

Critic target(PPO): own returns only

$$\rho_{i,t}(\theta_i) = \pi_{\theta_i}(a_{i,t} \mid h_{i,t} / \pi_{\theta_i^{old}}(a_{i,t} \mid h_{i,t}),$$
$$L_i^{\pi} = \mathbb{E}_t[\min\{\rho_{i,t}, \tilde{A}_{i,t}, \mathrm{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)\tilde{A}_{i,t}\}]$$

Simulation Results: Setup

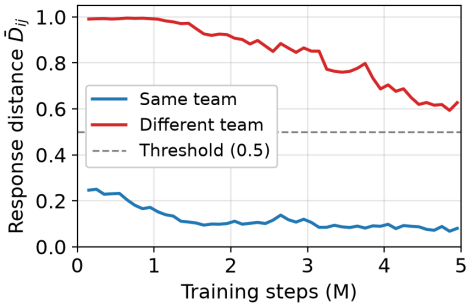
- **Environment**: Multi-team RWARE, 11×10 grid, 6 agents in 2 teams (3+3), team goals G0 and G1 at the bottom corners
- **Reward**: Individual reward only (delivery, shelf return, pickup); no team reward and no team labels
- **Communication**: Only on contact, $\|p_{i,t} - p_{j,t}\|_\infty \leq 3$
- **Evaluation**
  - Offline replay of independent PPO experience (15 agent pairs)
  - 5M-step training run with the discovery rule (seed 1)
  - Complete PGCT branched from a common 1.024M-step checkpoint, +0.5M steps (seeds 1, 3, 5)
- **Parameters**: $r \geq 0.5$, $\beta = 0.35$, $\bar{D}_{ij} < 0.5$, $n_{ij} \geq 3$, $\eta = 0.5$



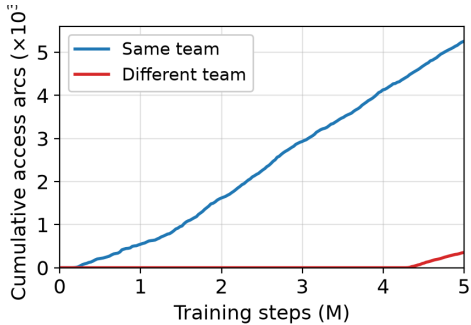
Rendered warehouse (G0, G1: team goals)

Simulation Results: Team Discovery during Training

Response distance



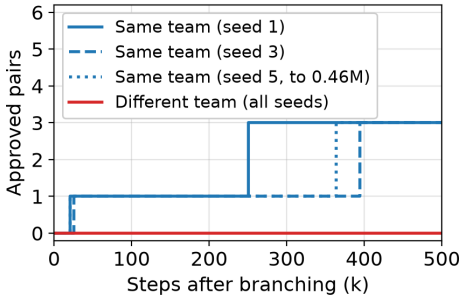
Peer critic access



- Observation 1: Teams separate as soon as agents mature (0.1–0.25M: 0.25 vs. 0.99)
- Observation 2: 93.7% of peer access went to true teammates (5,252 vs. 352 arcs); the first different-team approval appears at 4.28M
- Key message: Individual reward responses identify teammates without team labels

Simulation Results: Online Discovery with PGCT

Approved pairs after branching



Discovery per seed

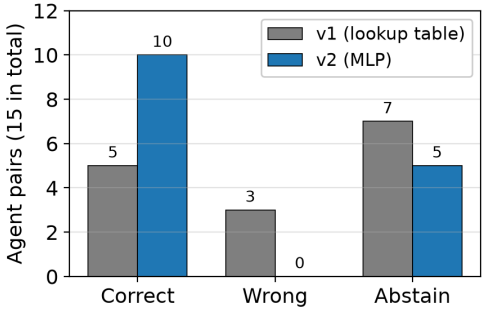
Seed	1st approval	Same	Diff.	Donor arcs
1	20.5k	3 / 6	**0**	1,920
3	25.6k	3 / 6	**0**	1,350
5†	20.5k	3 / 6	**0**	1,258

Same / Diff.: approved same-team / different-team pairs.
Donor arcs: all to true teammates. †Up to 0.46M steps.

- Observation 1: No different-team pair was approved in any seed
- Observation 2: The 3 approved pairs are all teammate pairs of mature agents; agents without own reward events abstain
- Key message: The complete method uses only teammates' critics

Simulation Results: Quantitative Comparison

Offline replay: decisions on 15 pairs



- **v1**: Lookup table of the mean own reward over identical public transitions
- **v2**: MLP reward predictor $\hat{r}_i(q; \psi_i)$ (proposed)
- **Accuracy**: v1 5/8 (62.5%), v2 10/10 (100%) of decided pairs
- **Abstain**: All 5 v2 abstentions involve the agent that never learned to deliver

- Key message: v1 rejected 3 teammate pairs; v2 decided more pairs without error

Conclusions

- **Summary**: PGCT discovers latent teammates online in fully decentralized MARL from individual reward responses on common event probes
- **Main contribution**: Reward-response team discovery that abstains without enough evidence, and selective use of teammate critics through Max-GAE
- **Validation**: Teammates separate early (0.25 vs. 0.99), and the complete method approved no different-team pair
- **Future work**: Learning-speed comparison with independent PPO and true-team critics, and support for agents without reward events

Thank You

Q&A